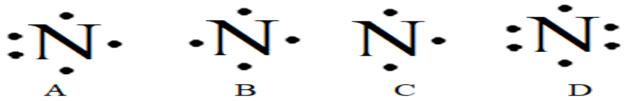
Chapter 8

1) The lewis symbol of nitrogen atom is: A



2) The number of valence electrons in Ca^{2+} is:

A) 8 B) zero C) 1 D) 2

3) Which of the following atoms gains three electrons to achieve octet rule is soluble in water?

A) Cl B) S C) P D) C

4) Which of the following atoms the lowest radius?

A) Si B) C C) P D) N

5) Which of the following has the highest ionic radius?

A) P^{3-} B) S^{2-} C) K^{+} D) Ca^{2+}

6) Which of the following ionic compounds has the highest lattice energy?

A) $AlCl_3$ B) LiF C) NaCl D) MgO E) CaO

7) Which of the following atoms the lowest electronegativity?

A) S B) Cl C) P D) Se E) Br

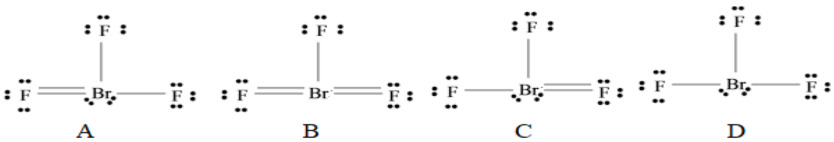
8) Which of the following bonds is the most polar?

A) N-O B) P-O C) P-S D) S-O

9) Which of the following diatomic ions forms triple covalent bond?

A) PO^{-} B) PO^{+} C) BrO^{+} D) BrO^{-}

10) The correct lewis structure of BrF_3 is: D



11) The number of bonding and nonbonding pairs of electrons around Br atom in BrO_3^{-} ion respectively are:

A) 3, 1 B) 2, 3 C) 4, 0 D) 2, 2

12) Which of the following ions include resonance structure?

A) SO_3^{2-} B) PO_3^{-} C) ClO_3^{-} D) ICl_3

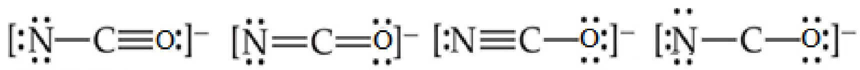
13) The formal charge of N in NO_2^- is:

A) zero B) -1 C) +1 D) -2

14) How many resonance structures are there in PO_3^- ion?

A) 2 B) No resonance C) 3 D) 4

15) The most preferable lewis structure of NCO^- ion is: C



A B C D

16) Which of the following molecules is polar?

A) XeF_2 B) PF_3 C) SF_4 D) PO_3^-

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17) In which of the following molecules (ions) the central atom violates octet rule.

A) NF_3 B) PCl_3 C) PI_3 D) $BrCl_3$

17) Which of the following Lewis structures is never to be exist.

A) ICl_5 B) FCl_3 C) PCl_3 D) SCl_4

18) Calculate the bond enthalpy of H-H bond: $N_2H_{4\,(g)} \longrightarrow N_{2\,(g)} + 2\,H_{2\,(g)}$ $\Delta H = -86$ kJ

If the bond enthalpies of N-N =163 kJ/mol, N≡N = 941 kJ/mol, N-H =391 kJ/mol

A) 872 B) 218 C) 436 D) 86

19) In which of the following molecules (ions) the central atom is less than octet rule.

A) BF_4^- B) SF_6 C) CCl_4 D) CCl_3^+

20) Which of the following atoms is never to found with more than octet of valence electrons around it?

A) N B) S C) Cl D) P

Chapter 9

1) The number of bonding domain and nonbonding domain around As atom in AsO_2^- ion respectively are:

A) 1, 2 B) 2, 1 C) 3, 0 D) 0, 3

2) The electron domain geometry of PCl_3 molecule is:

A) Trigonal pyramidal B) tetrahedral C) bent D) T-shape

3) The molecular shape of SF_4 molecule is:

A) Octahedral B) square planar C) see-saw D) T-shape

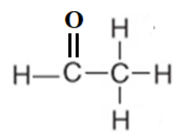
4) The VSEPR formula of ICl_3 molecule is:

A) AB_2E_3 B) AB_5 C) AB_3E_2 D) AB_4E

5) The bond angle in XeO_2 molecule is:

A) 90˚ B) 180˚ C) 109.5˚ D) 120˚

6) The bond angle (H-C-H) in the following molecule is:



A) 90˚ B) 180˚

C) 109.5˚ D) 120˚

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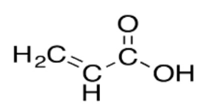
7) The hybridization of As atom in AsO_3 molecule is:

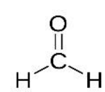
A) sp^3 B) sp^3d C) sp^2 D) sp^3d^2

8) How many σ and π bonds respectively are in the following molecule?



A) 8, 2 B) 2, 2 C) 2, 8 D) 8, 8

9) The C-O bond in the following molecule results from the overlap of:



A) sp (C) with sp (O) B) sp^2 (C) with sp^2 (O)

C) sp (C) with sp^2 (O) D) sp^2 (C) with sp (O)

10) Which of the following diatomic molecule is paramagnetic?

A) Be_2 B) B_2 C) N_2 D) F_2

11) Which of the following ions has the highest bond order?

A) O_2^{2-} B) O_2^+ C) O_2^{2+} D) O_2^-

12) the bond order of C_2^{2-} is:

A) 3 B) 2 C) 2.5 D) 1.5

13) Which of the following is diamagnetic?

A) O_2 B) O_2^{2-} C) O_2^{+} D) O_2^{-}

14) Which of the following molecules (ions) is non-polar?

A) CH_3Cl B) SO_2 C) ICl_3 D) NO_3^{-} E) SF_6

15) The true statement of the following is:

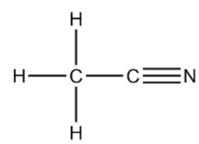
A) π bond results from overlap of tow s orbitals.

B) σ bond results from overlap of tow s orbitals only.

C) Generally, π bonds are weaker than σ bonds.

D) π bond results from overlap of s orbital with p orbital.

8) How many σ and π bonds respectively are in the following molecule?



A) 4, 2 B) 5, 1 C) 1, 5

D) 5, 2 E) 2, 5

Chapter 10

1) The False statement of the following is:

A) Gases expand spontaneously to fill its container

B) Gases are highly compressible

C) Gases have fixed volume and no fixed shape

D) Gases form homogeneous mixtures with each other

2) The atmospheric pressure in Irbid was recorded 715 mmHg. The pressure in kilopascal (kpas) is:

A) 0.94 B) 95325 C) 95.325 D) 715

3) Which of the following represent the lowest pressure:

A) 172235 Pa B) 114.427 kPa C) 340 torr D) 0.65 atm

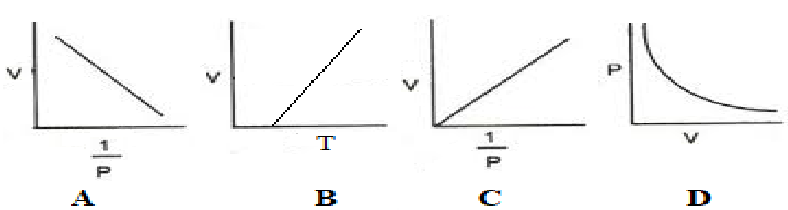
4) An open-end manometer containing mercury is connected to a container of gas. The mercury in the arm attached to the gas is 1.54 cm higher than that in the arm open to atmosphere. What is the pressure of gas if atmospheric pressure = 0.94 atm?

A) 0.942 B) 0.938 atm C) 0.96 atm D) 0.92 atm

5) Which of the following represent the lowest pressure: B

$V \propto \frac{1}{T}$ (A) $V \propto T$ (B) $V \propto n$ (C) $V \propto P$ (D) $V \propto \frac{1}{P}$ (E)

6) Which of the following graphs represents Boyle`s law? C



7) Which of the following processes will decrease the density of gas?

A) The gas is heated in a constant-volume container

B) Additional gas is added to a constant volume container

C) The gas expands at constant temperature

D) The gas is compressed at constant temperature

8) How many grams of O_2 (MM=32 g/mol) are there in 574 cm^3 vessel at STP. (R=0.0821 atm.L.$K^{-1}mol^{-1}$)

A) 0.28 g B) 0.82 g C) 0.67 g D) 0.76 g E) 0.574 g

9) Calculate the pressure (atm) of 0.57 g CO_2 (MM=44 g/mol) in 250 cm^3 vessel at 31 ˚C. (R=0.0821 atm.L.$K^{-1}mol^{-1}$)

A) 1.3 B) 1.6 C) 0.85 D) 0.57 E) 1.0

10) The volume of N_2 was recorded 213 cm^3 at STP and 264 cm^3 at 25 ˚C. The pressure of N_2 (mmHg) at 25 ˚C is:

A) 760 B) 640 C) 669 D) 860 E) 180

11) The volume of CH_4 gas at 30 ˚C is half its volume at 139 ˚C. If the pressure of CH_4 at 139 ˚C equals 0.17 atm, calculate the pressure at 30 ˚C.

A) 0.17 atm B) 0.085 atm C) 0.34 atm D) 0.25 atm E) 0.30

12) The density of NO_2 at STP is: (R=0.0821 atm.L.$K^{-1}mol^{-1}$, MM= 46 g/mol)

A) 2.05 g/mL B) 0.205 g/L C) 2.05 g/L D) 4.2 g/L E) 4.2 g/mL

13) The density of C_2H_6 gas (MM=30 g/mol) is three times the density of gas Y at the same temperature and pressure, the molar mass of gas Y is:

A) 30 g/mol B) 90 g/mol C) 10 g/mol D) 60 g/mol E) 15 g/mol

14) The density of gas X was measured 1.15 g/L at 25 ˚C and 750.9 torr, gas X could be:

A) C_2H_6 B) C_2H_4 C) O_2 D) CH_4

15) The volume (mL) of O_2 that could be evolved through the decomposition of 1.54 g $KClO_3$ (MM=122.6 g/mol) at 22 ˚C and 1.64 atm is: (R=0.0821 atm.L.$K^{-1}mol^{-1}$)

$2\ KClO_{3\,(s)} \longrightarrow 2\ KCl_{\,(s)} + 3\ O_{2\,(g)}$

A) 186 L B) 164 L C) 278 L D) 124 L E) 324 L

16) A 14.4 L of SO_2 was evolved through the reaction of 0.500 mole SO with O_2 at 25 ˚C, the pressure of SO_2 (mmHg) is:

$SO_{(g)} + 1/2\ O_{2\,(g)} \longrightarrow SO_{2\,(g)}$

A) 646 B) 323 C) 1292 D) 162

17) Calculate the total pressure (atm) of a mixture made by mixing of 3.60 g of O_2 (32 g/mol) with 6.70 g N_2 (28 g/mol) in 12 L container at 20 ˚C (R=0.0821 atm.L.$K^{-1}mol^{-1}$).

A) 0.226 atm B) 0.479 atm C) 0.253 atm D) 0.705 atm

18) A mixture composed of 3.5 g of O_2 (32 g/mol) and 8.3 g of unknown gas, the total pressure of this mixture equals 1.6 atm at 20 ˚C in 3.6 L container. The molar mass of unknown gas is:

A) 32 g/mol B) 64 g/mol C) 128 g/mol D) 256 g/mol

19) A mixture composed of two gases A&B, if the total moles equal 4.30, the total pressure equal 1.60, and the mole fraction of B gas equal 0.280. Calculate the partial pressure of gas A.

A) 0.450 atm B) 1.20 C) 1.15 atm D) 3.10

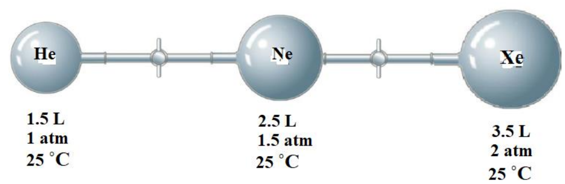
20) How many grams of $CaCO_3$ (MM=100 g/mol) could be decomposed to collect 352 mL CO_2 over water at 740 torr total pressure and 26 ˚C (the partial pressure of water at 26 ˚C=25 torr)?

$CaCO_{3\,(s)} \longrightarrow CaO_{\,(s)} + CO_{2\,(g)}$

A) 0.68 g B) 2.7 g C) 1.35 g D) 5.4 g

Q21-22

When the stopcock between the three containers in the following apparatus is opened and the gases allowed to mix



21) Calculate the partial pressure of Ne after mixing.

A) 0.93 atm B) 0.50 atm

C) 0.20 atm D) 1.5 atm

22) Calculate the mole fraction of Xe after mixing.

A) 0.57 B) 0.12 C) 0.31 D) 0.44

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23) Which of the following gases has the highest root mean square (rms) speed at 20 ˚C.

A) Xe B) Cl_2 C) CH_4 D) O_2 E) Ne

24) Calculate the root mean square (rms) speed for Ne (MM= 20 g/mol) at 25 ˚C.

A) 610 m/s B) 372 m/s C) 186 m/s D) 43 m/s

25) The root mean square (rms) speed for N_2 gas at 25 ˚C was 515 m/s. Calculate the (rms) speed for N_2 gas at 35 ˚C

A) 533 m/s B) 524 m/s C) 507 m/s D) 498 m/s

26) The rate of effusion of gas A is three times the rate of effusion of gas B (MM= 220 g/mol) at 25 ˚C and 1 atm. The molar mass of gas A is:

A) 739 g/mol B) 660 g/mol C) 24 g/mol D) 1980 g/mol

27) The two gases O_2 (MM= 32 g/mol) and CH_4 (MM= 16 g/mol) diffused the same distance, how long O_2 gas takes if CH_4 takes 20 second.

A) 40 seconds B) 10 seconds C) 28 seconds D) 14 seconds

28) How many centimeter the gas B (MM= 16 g/mol) should move to react with gas A (MM= 64 g/mol) if the two gases effused at the same time?

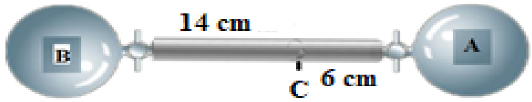
A) 5 cm B) 10 cm

C) 15 cm D) 7.5 cm E) 1.5 cm

29) When the two gases effused at the same time and moved as shown in the figure to react at point C. If the molar mass of gas A is 71 g/mol, the molar mass of gas B is:

A) 31 g/mol B) 13 g/mol

C) 166 g/mol D) 108 g/mol E) 384 g/mol



30) The volume (mL) of NH_3 produced when 20 mL of N_2 was added to 30 mL of H_2 is:

$N_{2\,(g)} + 3\,H_{2\,(g)} \longrightarrow 2\,NH_{3\,(g)}$

A) 50 mL B) 10 mL C) 20 mL D) 30 mL E) 40 mL

- For two gases at constant temperature ($T_1 = T_2$)
  $u_{rms} \propto 1/M$

$$\frac{r_1}{r_2} = \frac{u_1}{u_2} = \sqrt{\frac{3RT_1/M_1}{3RT_2/M_2}} = \sqrt{\frac{M_2}{M_1}}$$

- For two gases at different temperature

$$\frac{r_1}{r_2} = \frac{u_1}{u_2} = \sqrt{\frac{T_1/M_1}{T_2/M_2}} = \sqrt{\frac{T_1M_2}{T_2M_1}}$$

- For the same gas at two temperatures ($M_1 = M_2$)
  $u_{rms} \propto T$

$$\frac{r_1}{r_2} = \frac{u_1}{u_2} = \sqrt{\frac{3RT_1/M_1}{3RT_2/M_2}} = \sqrt{\frac{T_1}{T_2}}$$

- For two gases at constant temperature ($T_1 = T_2$):

$$\text{rate of effusion (r)} = \frac{\text{distance}}{\text{time}}$$

  - move at the same time ($Time_1 = Time_2$)

    Distance $\propto 1/M$

$$\frac{r_1}{r_2} = \frac{\text{distance}_1}{\text{distance}_2} = \sqrt{\frac{M_2}{M_1}}$$

  - move the same distance
    Time $\propto M$

$$\frac{r_1}{r_2} = \frac{\text{Time}_1}{\text{Time}_2} = \sqrt{\frac{M_1}{M_2}}$$